CHAPTER 5

Complex numbers and geometry

Angles and ratios of lengths, n-th roots of unity

Objectives. Interpret the modulus and an argument of $\frac{c-a}{b-a}$; prove that points are collinear, that lines are perpendicular, the nature of a triangle; determine sets of points; determine the n-th roots of unity and use them for regular polygons.

Prerequisites. Chapters 2, 3 and 4; oriented angles of vectors; special triangles.

Fluency

Fluency · No calculator, five minutes.

1. Give an argument of $-1+i$ and of $-3i$.
2. Write $\mathrm{e}^{2i\pi/3}$ in algebraic form.
3. If A and B have affixes a and b, what is the affix of $\overrightarrow{AB}$?
4. Compute $\dfrac{1+i}{1-i}$.
5. At most how many solutions does the equation $z^5 = 1$ have in $\mathbb{C}$?
6. What is the geometric effect of multiplication by $\mathrm{e}^{i\pi/3}$?

Solution · Fluency.

1. $\frac{3\pi}{4}$ and $-\frac{\pi}{2}$.
2. $-\frac{1}{2}+\frac{\sqrt{3}}{2}i$.
3. $b-a$.
4. $\frac{(1+i)^2}{2} = i$.
5. At most five (chapter 4).
6. A rotation with centre O and angle $\frac{\pi}{3}$.

Introductory activity

Activity 1 · The solutions of $z^n = 1$ 40 min .

1. Factorise $z^3 - 1$ by $z - 1$, then solve $z^3 = 1$ in $\mathbb{C}$. Let j be the solution with positive imaginary part.
2. Plot the three solutions in the complex plane. Compute the distances between them. Which figure do you obtain?
3. Write j in exponential form. Check that $j^2 = \bar{j}$ and that $1 + j + j^2 = 0$. Interpret the last equality with the centroid.
4. Solve $z^4 = 1$ and plot the solutions. Same question for $z^6 = 1$, using $z^6 - 1 = (z^3 - 1)(z^3 + 1)$.
5. State a conjecture about the solutions of $z^n = 1$: their number, their modulus, their arguments, the figure they form.

Observing is not proving: the properties conjectured here are established in the course.

Solution · Activity.

1. $z^3 - 1 = (z-1)\left(z^2 + z + 1\right)$; the quadratic has discriminant -3. Solutions: 1, $j = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ and $\bar{j} = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$.
2. The three distances equal $\sqrt{3}$: an equilateral triangle inscribed in the unit circle.
3. $j = \mathrm{e}^{2i\pi/3}$, $j^2 = \mathrm{e}^{4i\pi/3} = \mathrm{e}^{-2i\pi/3} = \bar{j}$, and $1 + j + j^2 = 1 - \frac{1}{2} - \frac{1}{2} + i\left(\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2}\right) = 0$. The affix of the centroid, $\frac{1+j+j^2}{3}$, is zero: it is O.
4. $z^4 = 1$: 1, i, -1, $-i$, vertices of a square. $z^6 = 1$: the solutions of $z^3 = 1$ and those of $z^3 = -1$, that is -1, $\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$; we obtain a regular hexagon.
5. Conjecture: n solutions, all of modulus 1, with arguments $\frac{2k\pi}{n}$; they are the vertices of a regular polygon with n sides inscribed in the unit circle, one of whose vertices is 1.

Course notes

Angles and lengths

Property · Interpretation of $\frac{c-a}{b-a}$. Let A, B, C be three points with affixes a, b, c, with $A \neq B$ and $A \neq C$. Then

$$\left|\frac{c-a}{b-a}\right| = \frac{AC}{AB} \qquad \text{and} \qquad \arg\left(\frac{c-a}{b-a}\right) \equiv \left(\overrightarrow{AB}, \overrightarrow{AC}\right) \pmod{2\pi}.$$

Proof. The modulus of a quotient is the quotient of the moduli: $|c-a| = AC$ and $|b-a| = AB$. For the argument, $\arg(b-a) \equiv \left(\overrightarrow{u}, \overrightarrow{AB}\right)$ and $\arg(c-a) \equiv \left(\overrightarrow{u}, \overrightarrow{AC}\right)$, so by Chasles' relation $\arg\frac{c-a}{b-a} \equiv \left(\overrightarrow{u}, \overrightarrow{AC}\right) - \left(\overrightarrow{u}, \overrightarrow{AB}\right) \equiv \left(\overrightarrow{AB}, \overrightarrow{AC}\right) \pmod{2\pi}$. □

Property · Collinearity, orthogonality. With the same notation, let $q = \frac{c-a}{b-a}$.

1. A, B, C are collinear if and only if q is real.
2. The lines (AB) and (AC) are perpendicular if and only if q is purely imaginary.
3. The triangle ABC is right-angled and isosceles at A if and only if $q = i$ or $q = -i$; it is equilateral if and only if $q = \mathrm{e}^{i\pi/3}$ or $q = \mathrm{e}^{-i\pi/3}$.

Proof. q is real if and only if its argument is 0 or π modulo 2π, that is if $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are collinear. q is purely imaginary (nonzero) if and only if its argument is $\pm\frac{\pi}{2}$. For 3, the previous theorem gives $|q| = \frac{AC}{AB}$ and $\arg q \equiv \left(\overrightarrow{AB}, \overrightarrow{AC}\right) \pmod{2\pi}$. Hence $q = \pm i$ if and only if $AC = AB$ and $\left(\overrightarrow{AB}, \overrightarrow{AC}\right) \equiv \pm\frac{\pi}{2}$, which characterises a triangle that is right-angled and isosceles at A. Likewise, $q = \mathrm{e}^{\pm i\pi/3}$ if and only if $AC = AB$ and $\left(\overrightarrow{AB}, \overrightarrow{AC}\right) \equiv \pm\frac{\pi}{3}$. A triangle that is isosceles at A with angle A measuring $\frac{\pi}{3}$ has two other equal angles, each measuring $\frac{1}{2}\left(\pi - \frac{\pi}{3}\right) = \frac{\pi}{3}$: its three angles are equal, so it is equilateral. Conversely, an equilateral triangle satisfies $AC = AB$ and its angle at A measures $\frac{\pi}{3}$. □

Example 1 · Constructing a square. Let A and B have affixes a and b. The direct square $ABCD$ has vertices $c = b + i(b-a)$ and $d = a + i(b-a)$: we go from $\overrightarrow{AB}$ to $\overrightarrow{BC}$ by a direct quarter turn. With $a = 0$ and $b = 1$: $c = 1 + i$ and $d = i$.

n-th roots of unity

Definition · n-th root of unity. Let $n \geq 1$ be an integer. An *n-th root of unity* is any complex number z such that $z^n = 1$. Their set is written $\mathbb{U}_n$.

Property · Determination of $\mathbb{U}_n$. For every integer $n \geq 1$, $\mathbb{U}_n$ has exactly n elements:

$$\mathbb{U}_n = \left\{\mathrm{e}^{2ik\pi/n} \;:\; k \in \{0, \ldots, n-1\}\right\}.$$

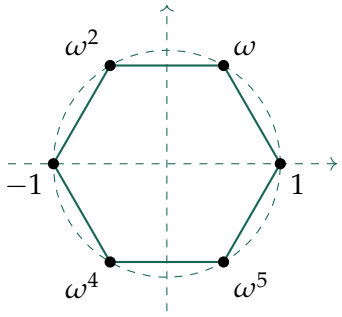
Proof. Let $z \in \mathbb{U}_n$. Then $|z|^n = |z^n| = 1$ with $|z| \geq 0$, so $|z| = 1$ and $z = \mathrm{e}^{i\theta}$ for some real number θ. The equality $\mathrm{e}^{in\theta} = 1$ is equivalent to $n\theta \equiv 0 \pmod{2\pi}$, that is $\theta = \frac{2k\pi}{n}$ with k an integer. Conversely, $\left(\mathrm{e}^{2ik\pi/n}\right)^n = \mathrm{e}^{2ik\pi} = 1$.

Two integers k and k' give the same number if and only if $\frac{2(k-k')\pi}{n}$ is a multiple of 2π, that is $k \equiv k' \pmod{n}$. Every integer k has the same remainder as an integer of $\{0, \ldots, n-1\}$, and these n integers have distinct remainders: $\mathbb{U}_n$ has exactly n elements. □

Key point · Geometric interpretation. For $n \geq 3$, the images of the elements of $\mathbb{U}_n$ are the vertices of a regular polygon with n sides inscribed in the unit circle, one of whose vertices is the point with affix 1. With $\omega = e^{2i\pi/n}$, we have $\mathbb{U}_n = \{1, \omega, \omega^2, \ldots, \omega^{n-1}\}$. Cases to know: $\mathbb{U}_2 = \{1, -1\}$, $\mathbb{U}_3 = \{1, j, j^2\}$ with $j = e^{2i\pi/3}$, $\mathbb{U}_4 = \{1, i, -1, -i\}$.

Property · Sum of the n-th roots. For every integer $n \geq 2$, the sum of the n-th roots of unity is zero.

Proof. With $\omega = e^{2i\pi/n} \neq 1$ (since $n \geq 2$): $1 + \omega + \cdots + \omega^{n-1} = \frac{1-\omega^n}{1-\omega} = 0$. □



The sixth roots of unity form a regular hexagon.

Exercises

Exercise 5.1 · Collinearity and orthogonality. Consider the points A, B, C, D with affixes $a = 1+i$, $b = 3+2i$, $c = -1$ and $d = 2-i$.

1. Compute $\dfrac{c-a}{b-a}$. What can be deduced?
2. Compute $\dfrac{d-a}{b-a}$. What can be deduced about the triangle ABD?

Solution.

1. $\dfrac{-2-i}{2+i} = -1$, real: A, B, C are collinear (A is even the midpoint of $[BC]$).
2. $\dfrac{1-2i}{2+i} = \dfrac{(1-2i)(2-i)}{5} = \dfrac{-5i}{5} = -i$: $AD = AB$ and $(AB) \perp (AD)$. The triangle ABD is right-angled and isosceles at A.

Exercise 5.2 · Nature of triangles. Determine the nature of the triangle ABC in each case:

1. $a = 0$, $b = 2$, $c = 1 + i\sqrt{3}$;
2. $a = 1$, $b = 2+i$, $c = i$.

Solution.

1. $\dfrac{c-a}{b-a} = \dfrac{1+i\sqrt{3}}{2} = \mathrm{e}^{i\pi/3}$: equilateral triangle.
2. $\dfrac{c-a}{b-a} = \dfrac{-1+i}{1+i} = \dfrac{(-1+i)(1-i)}{2} = \dfrac{2i}{2} = i$: triangle right-angled and isosceles at A.

Exercise 5.3 · Sixth roots. Determine the elements of $\mathbb{U}_6$ in exponential form, then in algebraic form. Check that their sum is zero.

Solution. $\mathrm{e}^{ik\pi/3}$ for $k \in \{0,\dots,5\}$: 1, $\frac{1}{2} + \frac{\sqrt{3}}{2}i$, $-\frac{1}{2} + \frac{\sqrt{3}}{2}i$, -1, $-\frac{1}{2} - \frac{\sqrt{3}}{2}i$, $\frac{1}{2} - \frac{\sqrt{3}}{2}i$. The real parts add up to $1 + \frac{1}{2} - \frac{1}{2} - 1 - \frac{1}{2} + \frac{1}{2} = 0$, and the imaginary parts cancel in pairs.

Exercise 5.4 · Cube roots of a complex number. Solve in $\mathbb{C}$ the equation $z^3 = 8i$. Write $z = r\mathrm{e}^{i\theta}$ and give the solutions in algebraic form.

Solution. $8i = 8\mathrm{e}^{i\pi/2}$. The equation is equivalent to $r^3 = 8$ and $3\theta \equiv \frac{\pi}{2} \pmod{2\pi}$, that is $r = 2$ and $\theta = \frac{\pi}{6} + \frac{2k\pi}{3}$. The solutions are $2\mathrm{e}^{i\pi/6} = \sqrt{3} + i$, $2\mathrm{e}^{5i\pi/6} = -\sqrt{3} + i$ and $2\mathrm{e}^{3i\pi/2} = -2i$. Check: $(-2i)^3 = -8i^3 = 8i$.

Exercise 5.5 · Completing a square. Let $A(1+2i)$ and $B(4+3i)$. Determine the affixes of the points C and D such that $ABCD$ is a direct square. Check that the diagonals have the same midpoint.

Solution. $b-a=3+i$ and $i(b-a)=-1+3i$. So $c=b+i(b-a)=3+6i$ and $d=a+i(b-a)=5i$. The midpoint of $[AC]$ has affix $\frac{4+8i}{2}=2+4i$, as does that of $[BD]$.

Exercise 5.6 · Sets of points. Let A and B be the points with affixes $2i$ and -1. To every point M with affix $z\neq -1$, we associate $Z=\dfrac{z-2i}{z+1}$.

1. Interpret geometrically the modulus and an argument of Z when $z\neq 2i$.
2. Determine the set of points M such that Z is real.
3. Determine the set of points M such that Z is purely imaginary and nonzero.
4. Determine the set of points M such that $|Z|=1$.

Solution.

1. $|Z|=\frac{MA}{MB}$ and $\arg Z\equiv\left(\overrightarrow{MB},\overrightarrow{MA}\right)\pmod{2\pi}$.
2. $Z=0$ for $M=A$. Otherwise, Z real and nonzero is equivalent to $\overrightarrow{MB}$ and $\overrightarrow{MA}$ collinear, that is $M\in(AB)$. The set is the line (AB) without B.
3. Z purely imaginary and nonzero is equivalent to $M\neq A$, $M\neq B$ and $(MA)\perp(MB)$: it is the circle with diameter $[AB]$ without A and B. Its centre has affix $\frac{-1+2i}{2}$ and its radius is $\frac{\sqrt{5}}{2}$.
4. $MA=MB$: the perpendicular bisector of $[AB]$.

Exercise 5.7 · A sum of cosines.

1. Let $n\geq 2$. Deduce from the course that $\displaystyle\sum_{k=0}^{n-1}\cos\frac{2k\pi}{n}=0$.
2. Write this equality for $n=5$ and prove that $1+2\cos\frac{2\pi}{5}+2\cos\frac{4\pi}{5}=0$.

Solution.

1. The sum of the elements of $\mathbb{U}_n$ is zero; so is its real part.
2. $\cos\frac{6\pi}{5}=\cos\left(2\pi-\frac{4\pi}{5}\right)=\cos\frac{4\pi}{5}$ and $\cos\frac{8\pi}{5}=\cos\frac{2\pi}{5}$. The sum $\cos 0+\cos\frac{2\pi}{5}+\cos\frac{4\pi}{5}+\cos\frac{6\pi}{5}+\cos\frac{8\pi}{5}$ therefore equals $1+2\cos\frac{2\pi}{5}+2\cos\frac{4\pi}{5}=0$.

Problem

Problem · Constructing a regular pentagon with ruler and compass. Let $\omega = e^{2i\pi/5}$ and $u = \omega + \frac{1}{\omega}$.

1. Prove that $u = 2\cos\frac{2\pi}{5}$.
2. Prove that $1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$, then, dividing by ω^2, that $u^2 + u - 1 = 0$.
3. Deduce that $\cos\frac{2\pi}{5} = \dfrac{\sqrt{5}-1}{4}$.
4. In an orthonormal frame, draw the unit circle and place $I(-\frac{1}{2}, 0)$ and $J(0, 1)$.
   (a) Compute IJ.
   (b) The circle with centre I through J meets the positive half of the horizontal axis at K. Prove that the abscissa of K is $2\cos\frac{2\pi}{5}$.
   (c) Deduce a ruler-and-compass construction of the vertices of a regular pentagon inscribed in the unit circle.

Solution · Problem.

1. $|\omega| = 1$, so $\frac{1}{\omega} = \overline{\omega}$ and $u = \omega + \overline{\omega} = 2\,\mathrm{Re}(\omega) = 2\cos\frac{2\pi}{5}$.
2. It is the sum of the fifth roots of unity. Dividing by $\omega^2 \neq 0$: $\omega^{-2} + \omega^{-1} + 1 + \omega + \omega^2 = 0$. Now $u^2 = \omega^2 + 2 + \omega^{-2}$, so $u^2 + u - 1 = \left(\omega^2 + \omega^{-2}\right) + \left(\omega + \omega^{-1}\right) + 1 = 0$.
3. The roots of $X^2 + X - 1$ are $\frac{-1\pm\sqrt{5}}{2}$. Since $0 < \frac{2\pi}{5} < \frac{\pi}{2}$, $u > 0$: $u = \frac{\sqrt{5}-1}{2}$ and $\cos\frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$.
4. (a) $IJ = \sqrt{\frac{1}{4} + 1} = \frac{\sqrt{5}}{2}$.
   (b) K has abscissa $-\frac{1}{2} + \frac{\sqrt{5}}{2} = \frac{\sqrt{5}-1}{2} = u$.
   (c) Construct K, then the midpoint H of $[OK]$, whose abscissa is $\cos\frac{2\pi}{5}$. The perpendicular to the axis at H meets the unit circle at ω and $\overline{\omega} = \omega^4$. Marking off the length $|\omega - 1|$ with the compass from these points gives ω^2 and ω^3.

Teacher note · Status of the problem. The trigonometric values of $\frac{2\pi}{5}$ and the construction of the pentagon are a "possible problem" of the syllabus. Have the construction carried out with dynamic geometry software, then check that the five sides have the same length ($|\omega - 1| = 2\sin\frac{\pi}{5}$).